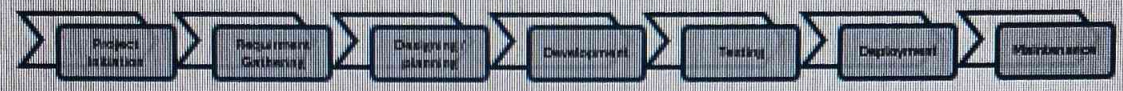
DAY - 01

ACTIVITY – 01

SOFTWARE DEVELOPMENT LIFECYCLE



1.PROJECT INITIATION

This is the starting point of any software project.

Purpose:

To decide whether the project should be started.

Activities include:

- Understanding the main problem or need
- Creating a high-level project idea
- Estimating time, cost, and resources
- Identifying stakeholders
- Checking if the project is feasible
- Getting approval to start the project

2.REQUIREMENT GATHERING

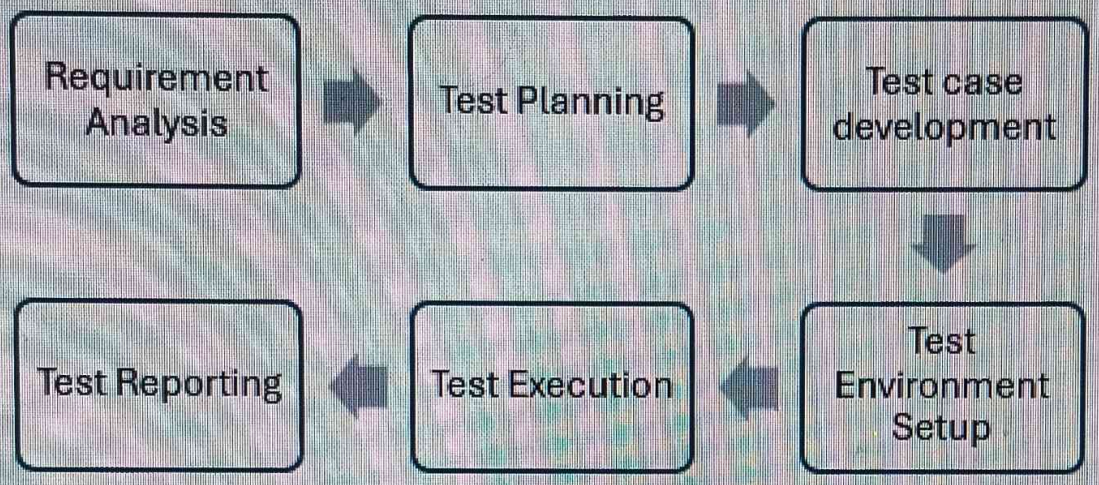
This is the first and most important phase.
The goal is to clearly understand what the customer wants.

Activities include:

- Meeting with clients/stakeholders
- Understanding features needed
- Identifying problems the software should solve

ACTIVITY – 02

SOFTWARE TESTING LIFE CYCLE



1. Requirement Analysis

- We understand what needs to be tested.
- Read the requirement document and identify key features.
- Clear any doubts with the team.

Example:
If the app has QR attendance, we understand how it should work.

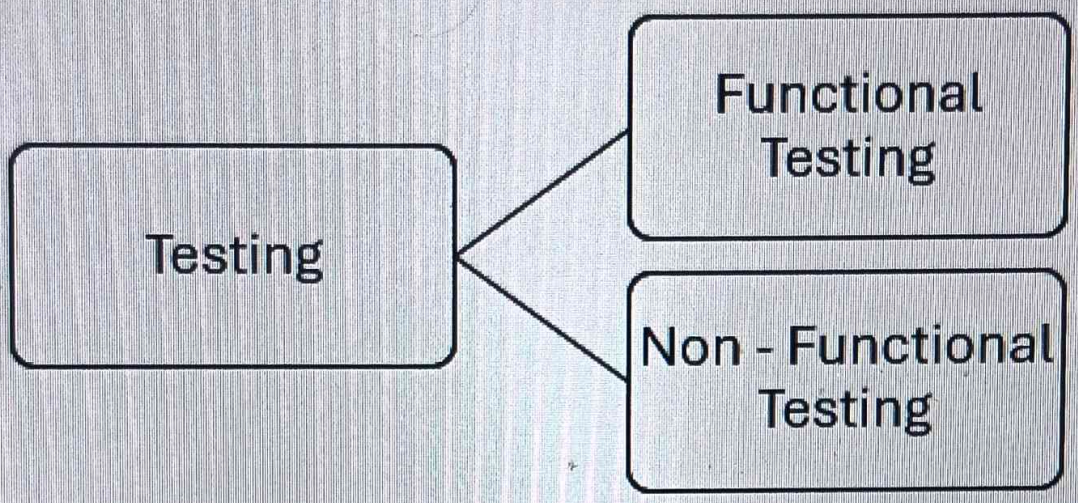
2. **Test Planning**

- We decide how testing will be done.
- Estimate time, number of testers, and resources needed.
- Prepare a test plan document.

Example:
Plan to test login, QR scan, reports, UI, performance, etc.

ACTIVITY – 03

TYPES OF TESTING



FUNCTIONAL TESTING

Functional Testing is a type of software testing that checks what the system does. It verifies that every function of the application works according to the requirement.

Examples:

- Login should work with correct username & password.
- Clicking Submit should save the data.
- QR code scan should mark attendance successfully.
- Clicking Download Report should generate the correct file.

TYPES OF FUNCTIONAL TESTING

1.UNIT TESTING

Unit Testing is a type of testing where individual components, small sections of code, or single functions are tested separately to ensure they work correctly on their own.

ACTIVITY – 04

PRINCIPLES OF TESTING

1. TESTING SHOWS PRESENCE OF DEFECTS

- Testing helps us find bugs, but it cannot prove the software has zero defects.
- Even after testing, some bugs may still exist.

Example:
You fix 10 bugs, but there may still be hidden bugs.

2. Exhaustive Testing is Impossible

- We cannot test everything (all inputs, all combinations).
- So we test important and risky areas first.

Example:
You cannot test all 1,000,000 login combinations; you test the main ones.

3. Early Testing Saves Time & Money

- Start testing **as early as possible**, even from requirement stage.
- Early bugs are cheaper to fix.

Example:
If requirement is wrong, fixing it early saves huge cost later.

4.Defects Cluster Together

- Most defects are usually found in few modules.
- These areas need more testing.

ACTIVITY – 05

BRD (BUSINESS REQUIREMENT DOCUMENT) & FS (FUNCTIONALITY SPECIFICATION DOCUMENT) FOR AMAZON WEBSITE

BUSINESS REQUIREMENT DOCUMENT

BRD_01

When a user opens Amazon.in, the Home Page should load first.

BRD_02

The Home Page should have a search bar to search for products.

BRD_03

The Home Page should show categories like Mobiles, Electronics, Fashion, Appliances, Grocery, Books, Beauty, etc.

BRD_04

Users should be able to sign in using their email or mobile number and password.

BRD_05

New users should be able to create an Amazon account using email or mobile number.

BRD_06

Users should be able to see product details like name, price, image, description, specifications, offers, and ratings.

BRD_07

Users should be able to Add products to the Cart.

BRD_08

Users should be able to change the quantity or remove items from the cart.

BRD_09